

## Jolly Roger

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## Re: Jolly Roger's Security Thread for Beginners

« Reply #85 on: March 13, 2014, 06:11:29 am »

- [Quote](#)

### **DENIABILITY, IDENTIFYING TAILS USERS, AND CAN YOU BE FORCED TO GIVE UP YOUR PASSWORDS?**

Quote from: OCDPolak

Hi JR,

For some reason I have seen a lot of information and discussion about privacy and anonymity but nothing at all about deniability, which to be honest concerns me that some people may think that because the NSA can't crack their password, everything is safe but people easily overestimate their ability to stand up to sanctions imposed by a court should the shit hit the fan...

I was wondering about the deniability problems with using Tails (or any of the security measures really). You have to assume that if you get

arrested and it goes to court, you will be compelled to give any of your passwords that they want. It's all well and good thinking that you won't give it to them, but when they sentence you to a \$1000 a day or simply jail until you tell them you will probably tell them your passwords...

With that in mind, is there any deniable way to use Tails (or at least deniable in some respects)? I used to run everything off a hidden volume in a Truecrypt memystick (which is supposed to be impossible to prove exists), is there an equivalent with LUKS?

Also, can your ISP or FBI differentiate between Tor and Tails through your internet usage?

Thanks for your time

Here are some things to consider.

Quote

Tails makes it clear that you are using Tor and probably Tails

Your Internet Service Provider (ISP) or your local network administrator can see that you're connecting to a Tor relay, and not a normal web server for example. Using Tor bridges in certain conditions can help you hide the fact that you are using Tor.

The destination server that you are contacting through Tor can know whether your communication comes out from a Tor exit node by consulting the publicly available list of exit nodes that might contact it. For example using the Tor Bulk Exit List tool of the Tor Project.

So using Tails doesn't make you look like any random Internet user. The anonymity provided by Tor and Tails works by trying to make all of their users look the same so it's not possible to identify who is who amongst them.

**<https://tails.boum.org/doc/about/warning/index.en.html#index2h1>**

Quote

In this context, the term fingerprint refers to what is specific to Tails in the way it behaves on Internet. This can be used to determine whether a particular user is using Tails or not.

As explained on our warning page, when using Tails it is possible to know that you are using Tor. But Tails tries to make it as difficult as

possible to distinguish Tails users from other Tor users, especially Tor Browser Bundle (TBB) users. If it is possible to determine whether you are a Tails user or a TBB user, this provides more information about you and in consequence reduces your anonymity.

This section explains some issues regarding the fingerprint of Tails and how this could be used to identify you as a Tails user.  
For the websites that you are visiting

The websites that you are visiting can retrieve a lot of information about your browser. That information can include its name and version, window size, list of available extensions, timezone, available fonts, etc.

To make it difficult to distinguish Tails users from TBB users, the Tor browser tries to provide the same information as the TBB in order to have similar fingerprints.

See the fingerprint section of know issues page for a list of known differences between the fingerprints of the Tor browser and the TBB.

Apart from that, some of the extensions included in Tor browser are different than the ones included in the TBB. More sophisticated attacks can use those differences to distinguish Tails user from TBB users.

For example, Tails includes Adblock Plus which removes advertisements. If an attacker can determine that you are not downloading the advertisements that are included in a webpage, that could help identify you as a Tails user.

For the moment, you should consider that no special care is taken regarding the fingerprint of the Unsafe Browser.  
For your ISP or local network administrator

Tor bridges are most of the time a good way of hiding the fact that you are connecting to Tor to a local observer. If this is important for you, read our documentation about bridge mode.

A Tails system is almost exclusively generating Tor activity on the network. Usually TBB users also have network activity outside of Tor, either from another web browser or other applications. So the proportion of Tor activity could be used to determine whether a user is using Tails or the TBB. If you are sharing your Internet connection with other users that are not using Tails it is probably harder for your ISP to determine whether a single user is generating only Tor traffic and so maybe using Tails.

Tails do not use the entry guards mechanism of Tor. With the entry guard mechanism, a Tor user always uses the same few relays as first hops. As Tails does not store any Tor information between separate working sessions, it does not store the entry guards information either. This behaviour could be used to distinguish Tails users from TBB users across several working sessions.

When starting, Tails synchronizes the system clock to make sure it is accurate. While doing this, if the time is set too much in the past or in the future, Tor is shut down and started again. This behavior could be used to distinguish Tails from TBB users, especially this happens every time Tails starts.

**<https://tails.boum.org/doc/about/fingerprint/index.en.html>**

Read those pages directly as they have links to other articles on them as well.

Here is another little trick I know of. Never keep a password you can remember. You cannot give up a password you do not know. Perhaps you have a little piece of paper with your password on it that you swallow the second the cops come in. A long password that you could never remember.

Another thing you can say is, I wrote down my password on a piece of paper but the police must have destroyed the piece of paper when they raided my home. Check out the below quote from an article.

Quote

Dubois said that, in addition, his client may not be able to decrypt the laptop for any number of reasons. "If that's the case, then we'll report that fact to the court, and **the law is fairly clear that people cannot be punished for failure to do things they are unable to do,**" he said.

[http://news.cnet.com/8301-31921\\_3-57364330-281/judge-americans-can-be-forced-to-decrypt-their-laptops/](http://news.cnet.com/8301-31921_3-57364330-281/judge-americans-can-be-forced-to-decrypt-their-laptops/)

And in the case of whether or not you can be forced to give up a password is a matter of debate that has gone back and forth in court cases to date.

Quote

Many in the legal arena say the issue is a tricky -- and largely unsettled one.

A small number of courts have permitted it, but only when prosecutors can point to specifically what files they need and where they are located.

In the motion filed earlier this week, Assistant County Prosecutor Matthew Meyer stated the law is not clear.

[http://www.cleveland.com/court-justice/index.ssf/2014/03/bedford\\_judge\\_case\\_highlights.html](http://www.cleveland.com/court-justice/index.ssf/2014/03/bedford_judge_case_highlights.html)

And what about the charges for failing to do so?

Quote

disobeying a judge's order to hand over a password could result in contempt of court charges or being jailed.

[http://www.cleveland.com/court-justice/index.ssf/2014/03/bedford\\_judge\\_case\\_highlights.html](http://www.cleveland.com/court-justice/index.ssf/2014/03/bedford_judge_case_highlights.html)

And in the US, since most people busted will be extradited there anyways, treats contempt in the following way.

Quote

If a person is to be punished criminally, then the contempt **must be proven beyond a reasonable doubt**, but once the charge is proven, then punishment (such as a fine or, in more serious cases, imprisonment) is imposed unconditionally.

A court cannot maintain an order of contempt where the imposed party does not have the ability to comply with the underlying order. This claim when made by the imposed party is known as the "impossibility defense".

[https://en.wikipedia.org/wiki/Contempt\\_of\\_court#United\\_States](https://en.wikipedia.org/wiki/Contempt_of_court#United_States)

Furthermore.

Quote

“the government must prove the existence and location of the subpoenaed documents and possess independent evidence, other than compliance with the court order, for authenticating them” [1, p. 581]. In other words, law enforcement cannot simply go on a fishing expedition, hoping to turn up data that will be evidentiary [8]. They must be able to demonstrate the existence and likely location of specific documents.

[http://www.asis.org/Bulletin/Dec-13/DecJan14\\_Oltmann.html](http://www.asis.org/Bulletin/Dec-13/DecJan14_Oltmann.html)

In regards to two cases in which defendants were not forced to give up their passwords

Quote

United States v. Kirschner (2010): Kirschner was indicted for child pornography charges, and the government subpoenaed his encryption key to gain further evidence from his encrypted drive. In this case, the judge determined that requiring a defendant to supply his password would violate his right against self-incrimination.

United States v. Doe (2012): Doe was charged with child pornography. He refused to supply his decryption key and was found in contempt of court, then jailed. A judge then ruled that supplying his decryption key would be tantamount to self-incrimination, so Doe did not have to supply it.

[http://www.asis.org/Bulletin/Dec-13/DecJan14\\_Oltmann.html](http://www.asis.org/Bulletin/Dec-13/DecJan14_Oltmann.html)

The analysis of why they were not forced to give them up is below.

Quote

In contrast, law enforcement in the **Kirschner and Doe cases did not have prior evidence that illegal content was on their computers. In these cases, officers had suspicion of wrongdoing and were relying on the revelation of decryption keys to investigate and uncover evidence.** The court in Kirschner determined that sharing the key “would be testimonial because it would demonstrate knowledge of the password and access to the underlying computer files ...providing the password would reveal the contents of an arrestee’s mind by recalling the password” [5, pp. 1171-1172], [6]. Simply put, because the password was not written down (or already known to law enforcement) in Kirschner and Doe, and it existed only in their minds, compelling a defendant to reveal it would be self-incriminating testimony.

If law enforcement can describe the existence and location of evidence, they have a stronger case for requiring access; however, if they cannot demonstrate prior knowledge of the likely data, separate from a compelled revelation from a defendant, then law enforcement has a weaker position.

[http://www.asis.org/Bulletin/Dec-13/DecJan14\\_Oltmann.html](http://www.asis.org/Bulletin/Dec-13/DecJan14_Oltmann.html)

But when law enforcement was able to provide proof of existing evidence on an encrypted drive, courts were much more likely to demand decryption, such as in the following cases.

Quote

In re Boucher (2009): Boucher entered the United States from Canada. A border agent examined Boucher’s computer and found child pornography after Boucher supplied the password. The agent then shut down the computer and arrested Boucher. Shutting down the

computer triggered the encryption again, and prosecutors could no longer see or find the illegal images. Boucher was ordered by the courts to supply the password, but he invoked his Fifth Amendment privilege. The courts subsequently ruled he had to supply a decrypted copy of the drive's contents.

Commonwealth v. Hurst (2011): Hurst was charged with offenses related to inappropriate sexual relations with a minor. Police suspected incriminating evidence was on Hurst's cellphone, but he refused to supply the password. Before this case reached the court system, Hurst's wife supplied the password, and Hurst himself pled guilty.

United States v. Fricosu (2012): Fricosu was indicted for mortgage and real estate fraud. She refused to surrender the password (at one point saying she forgot the password) to encrypted files that, the government believed, would incriminate her. The court ordered her to supply a decrypted version of the hard drive, rather than her password. Subsequently, a co-defendant supplied the needed passwords.

And the analysis of the cases below.

Quote

Law enforcement saw evidence of criminal wrongdoing in the Pearson, Boucher, Hurst and Fricosu cases.

Both Pearson and Boucher voluntarily agreed to let law enforcement search their computers; during those searches, the officers saw evidence. It was only after the initial search that the question of encryption became relevant. In these cases, because the defendants had “permitted investigators to see at least some” of the evidence, this “sufficed to render the existence of all the illegal files a ‘foregone conclusion’” rather than testimonial evidence [8, p. 544]. Hurst had sent inappropriate messages to a minor, which were visible on the minor's phone. While the police sought confirmation of the transmission by searching Hurst's phone, they had sufficient evidence without that step. In the Fricosu case, police had recorded conversations between the defendant and her husband (a co-defendant) that revealed the existence and content of the sought-after documents.

[http://www.asis.org/Bulletin/Dec-13/DecJan14\\_Oltmann.html](http://www.asis.org/Bulletin/Dec-13/DecJan14_Oltmann.html)

- 1) It may be possible, to identify you as a Tails user, but it would take a lot of analysis to do so, and Tails is getting better at blending in with every update.
- 2) Think about what you could possibly be charged with, and think about whether or not it is more serious than a contempt charge. The longest sentence to date for contempt was 14 years, and this is almost unheard of. You are not likely to get this kind of charge against you, but if you do, would it be better than life in prison for whatever else you might be charged with?

Remember Sabu to LulzSec hacker? being charged with 112 years in prison for hacking? I think he would trade 14 years in prison for contempt over 112 years any day. I know I would.

3) Without the knowledge of incriminating evidence existing on your drives, you are less likely to be forced to decrypt your drives, and this even applies in child pornography cases as demonstrated above.

4) Maintain your right to remain silent, never keep anything on your computers that you do not have to.

5) Do not have a password you can remember. Or if you do, tell them you had it written down but it was misplaced or possibly damaged during the raid and you are unable to recall the password. Perhaps you are too traumatized from the even of having your face shoved into the floor to remember what happened during those 2 minutes?

Anyways, this is a lot of data to go through, so I will leave it at that and we can go from there. You just need to always follow best practices. Turn off your computer when you are not using it, encrypt everything, never tell anybody your passwords, never leave any evidence of the contents of your drives lying around (like notes or diary entries), and never admit having anything on your drives to anyone online, even under your pseudonym as that can be used against you in court.

Deny deny deny deny deny.

Hope this helps.

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[Logged](#)

They would live and die under it.

Jolly Roger's Security Thread For Beginners

<http://thehubaoydxrommh.onion/index.php?board=1.0>

BTC Donations: 1**JoLLy**5gwsMQUMavyqZeXwUoHBb693ZeYm